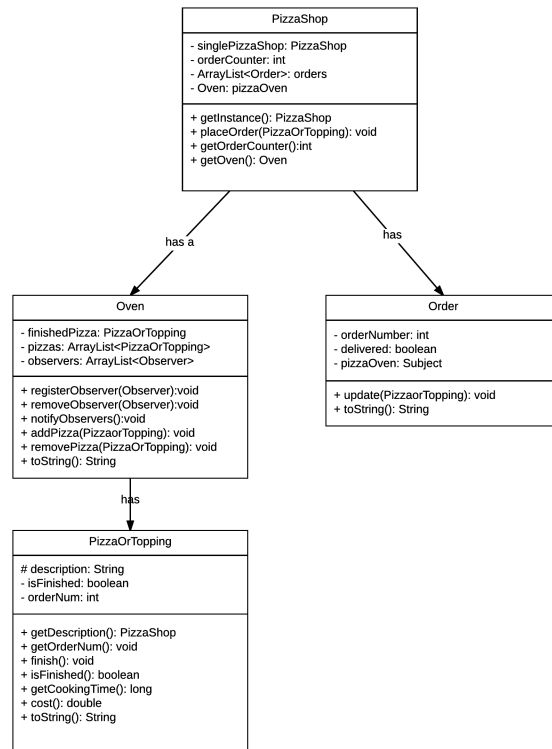


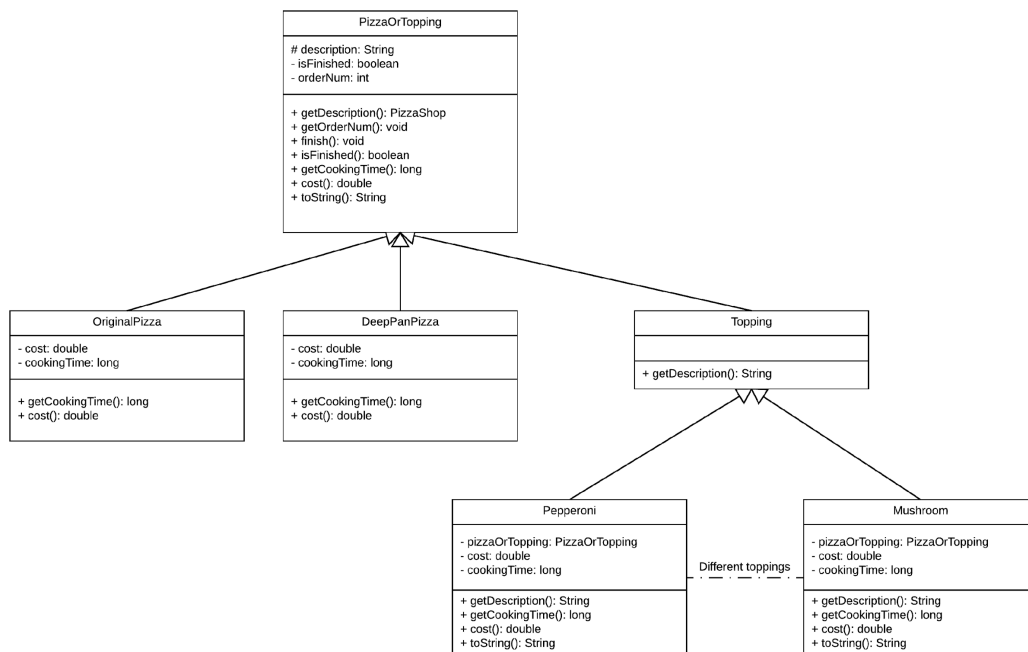
Programming Assignment #3 Extras & UML

The Assignment is divided into numerous java classes. To better understand how they connect you may take a look at the UML diagrams below.

These are the main classes of the app.



The pizza and toppings classes are shown here so you can see their relationships.



Hints

When implementing your singleton PizzaShop class you should ensure that it is well protected against threads.

When traversing the ArrayList you must be careful to use an appropriate loop;

- If any method within the loop will edit the ArrayList then you must use a regular for loop.
- If the loop will not edit the ArrayList then you can use a for-each loop or iterators.

If you are not careful then you may get a **ConcurrentModificationException**.

Do not change the Timer code. Simply insert it into the addPizza(...) method.

Using the Launchers

Launchers have been provided so that you can run your code. There are two launchers provided. To use a launcher, you just need to copy the file into the folder containing your java files.

If you use a package then you may have to add that to the launcher. This should be the only thing you need to change.

```
package pizza;
import java.util.InputMismatchException;

public class SimpleLauncher implements Observer{
```

1. SimpleLauncher:
 - a. This contains a simple text interface.
 - b. To use it you simply enter the selection via the keyboard for example.
 - c. To order an original pizza with mushrooms and extra cheese you can choose:
1, Enter, 1, Enter, 4, Enter, 5, Enter, 6, Enter.
 - d. When the pizza is finished, it will be displayed at the end.
2. GUILauncher:
 - a. This launcher creates a window with a single button.
 - b. Click the button to launch a pizza shop window.
 - c. When an order is made, the order is displayed on the right.
 - d. The order turns green when the pizza is finished.
 - e. You can launch multiple pizzashop windows but it will always be the same Pizza Shop {the order numbers are all the same count}